

Economic Theorizing

Economic theories that constitute the body of economic science are the result of scientific investigation of economic phenomena. Scientific method of investigation involves *observation of economic phenomena and collection and analysis of relevant facts with the purpose of establishing the relationship between the related economic variables*. When the relationship between the selected variables is established with a high degree of confidence, it is presented in the form of a theory or a hypothesis. This process is called *theorization or formulation of theory*.

An important element of scientific method of inquiry is *model building*. An analytical *model* is an abstraction of an economic phenomenon from the complex economic conditions of real life. It represents reality in a simplified form. An economic model is constructed in the form of mathematical equations specifying the relationship between the interrelated economic variables. Models are used to work out the implications of a theory, to deduce the consequences of the assumptions and to make predictions. *Economic variables* are measurable quantities, e.g., consumer goods, output, inputs, money, income, etc. The economic variables assumed to remain constant are called *parameters*.

In this section, we will briefly discuss the process of model building and theorization in economics in general.

Model Building and Economic Theorization. A scientific method of study or investigation consists of the following steps:

- ☐. Specifying the problem
- ☐. Formulating hypothesis
- ☐. Making assumptions
- ☐. Building a relevant model
- ☐. Collection and analysis of relevant data or facts
- ☐. Deducing the testable predictions from data analysis
- ☐. Testing the validity of predictions
- ☐. Formulation of the theory

The *first step* in scientific method of study is to specify the *problem*—the economic phenomenon chosen for the purpose of study. For example, the problem of a study may be specified as finding the effect of increase in the price of petrol on the demand for cars or the reasons for exorbitant increase in the price of onion in India, in December 2010 and its effect on food inflation.

The *second step* is to *formulate a hypothesis*. A hypothesis is a statement expressing the relationship between the cause and effect. A hypothesis may be expressed in the form of a statement, e.g., 'when price of petrol increases, demand for cars decreases'. Here, increase in petrol prices is the *cause* and decrease in the demand for cars is the *effect*. It may also be stated in the form of an equation, e.g., $Q_c = C - cP_p$ (where Q_c = number of cars demanded; C = cars demanded at zero price of petrol; $-c$ = a parameter showing the effect of change in petrol price on the demand for car; and P_p = price of petrol).

The *third step* is to make necessary *assumptions*. Assumptions are made to simplify the problem, to specify the variables and components of the model and to avoid the complexities that might arise due to the change in extraneous factors—factors operating outside the model. For instance, in our petrol-price and-car-demand example, some general assumptions are made as (i) people's income does not change significantly, (ii) car prices are not likely to change significantly, and (iii) there is no demonstration effect, etc. In general, in building an economic model, assumptions include (i) *behavioural or motivational assumptions* pertaining to the behaviour of the decision makers and their motivation or the objective that they set for themselves; (ii) *institutional assumptions* pertaining to the institutional set-up or market conditions under which the economic players (consumers and producers) seek to achieve their goals; (iii) *technological assumptions* relating to production technique; and (iv) *input-related assumptions*—those pertaining to the supply position of the inputs.

Once the assumptions are specified, the *fourth step* is to build a model for testing the hypothesis.

A model is constructed by specifying the relationship between the chosen variables. The relationship between the variables may be expressed in the form of mathematical equations or also in verbal terms.

The model so built forms the basis of data analysis and hypothesis testing.

The *fifth step* is to collect the relevant data and other facts related to the problem of study and their analysis according to the model. After data are collected, assembled and analysed, the next or the *fifth step* is to make the deduction/deductions, i.e., derive conclusions, about the relationship between the causal factor and its effect. In our example, the deduction

may be stated as 'when petrol price increases, demand for car decreases'. This deduction is the same as the hypothesis.

The *sixth step* is to test the validity of the model. The test of validity of a model is determined by its power to predict. In our example, the test of validity of the model is to find whether it can be used to predict the demand for car when the price of petrol increases (or decreases) by a certain percentage. If the model predicts the car demand fairly accurately. It is taken to be a valid model. Otherwise, it is an invalid model. In case a model turns out to be invalid, it needs to be modified by including other relevant and explanatory variables that influence the demand for car and put to test again. When the model is tested and retested and found to be valid, its outcome is stated in the form of a theory.

The *final step* is to state the findings of the study in the form of a theory. This leads to the formulation of a theory.